Supervised Graph Contrastive Learning for Gene Regulatory Networks

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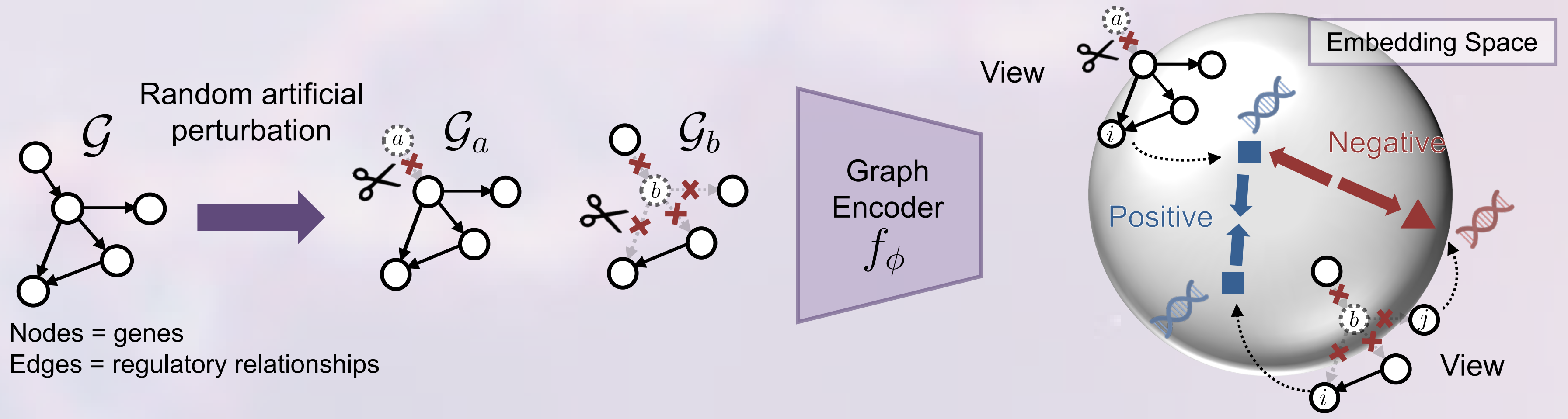
Background

- ✓ Learning effective gene regulatory networks (GRNs) representation is essential for biological tasks. ex.) Prediction of gene function and patient prognosis.
- ✓ Graph contrastive learning (GCL) has emerged as a powerful paradigm for learning graph representations.
  - The contrastive learning objective can be formulated as minimizing the KL divergence between the teacher distribution: p and the learned distribution: q_ϕ[1].



Nodes = genes
Edges = regulatory relationships

Node-level GCL objective

$$\mathrm{Loss}^{a,b}_{\mathrm{Node}} \triangleq \frac{1}{|\mathcal{V}|}\sum_{i\in\mathcal{V}} D_{\mathrm{KL}}\left(p(j\mid i)\,||\,q^{a,b}_\phi(j\mid i)\right)$$

$$\mathrm{Loss}_{\mathrm{Node}} \triangleq \mathbb{E}_{a,b\sim \mathrm{U}_\mathcal{A}}\left[\mathrm{Loss}^{a,b}_{\mathrm{Node}}\right]$$

Teacher distribution

$$p(j\mid i) \triangleq \delta_{ij}$$

Learned distribution

$$q^{a,b}_\phi(j\mid i) \triangleq \frac{\exp(\mathrm{sim}(z^a_i, z^b_j)/\tau_{\mathrm{n}})}{\sum_{k\in\mathcal{V}}\exp(\mathrm{sim}(z^a_i, z^b_k)/\tau_{\mathrm{n}})}$$

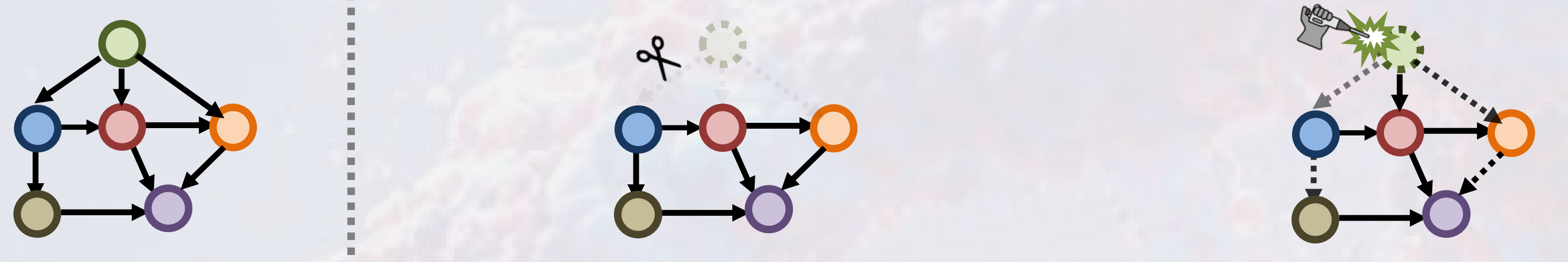
Notation

$\mathcal{V}, \mathcal{A}$: Sets of vertices or augmentation indices
$\mathcal{G}_a, \mathcal{H}_a$: Patient or experimental cell GRNs in view a
f_ϕ : Embedding mapping by GNN
$Z^a, Y^a \in \mathbb{R}^{|\mathcal{V}|\times d}$: The embedding of $\mathcal{G}_a$ or $\mathcal{H}_a$
$z^a_i \in \mathbb{R}^d$: The embedding of node i in view a
δ_{ij} : Kronecker delta
$\mathrm{sim}(\cdot,\cdot)$: Cosine similarity
$\mathrm{sim}_F(\cdot,\cdot)$ Frobenius cosine similarity
$\tau_{\mathrm{n}}, \tau_{\mathrm{a}}$: Temperature per node or augmentation
$\mathrm{U}_\mathcal{A}$: Uniform distribution on $\mathcal{A}$

Problem & Motivation

⚠ Artificial perturbations may alter biological meaning.

Original GRN | (1) Artificial perturbation (Data augmentation) | (2) Real experimental perturbation (Gene Knockdown)



(Baseline)

⚠ Removes key regulator; biologically misleading

✓ Biologically grounded; meaningful signal

Existing methods either rely on artificial perturbations or fail to leverage real perturbation-induced structural changes.

We aim to incorporate biologically meaningful perturbations into GCL to fully leverage perturbation effects for representation learning.

Method & Theory

Proposed method : **SupGCL** (**Sup**ervised **G**raph **C**ontrastive **L**earning)

- Using gene knockdown experiments as supervision for graph augmentation
- A unified KL objective for node- and augmentation-level learning

Full SupGCL objective

$$\mathrm{Loss}_{\mathrm{General}} \triangleq \frac{1}{|\mathcal{V}||\mathcal{A}|}\sum_{(i,a)\in\mathcal{V}\times\mathcal{A}} D_{\mathrm{KL}}\left(p(j,b\mid i,a)\,||\,q_\phi(j,b\mid i,a)\right)$$

$b\sim \mathrm{U}_\mathcal{A}$
$p(j\mid i)\triangleq \delta_{ij}$

$a,b\sim p_\phi(a,b)$

Existing method (Node-level GCL)

$$\mathrm{Loss}_{\mathrm{Node}} = \mathbb{E}_{b\sim\mathrm{U}_\mathcal{A}}\left[\frac{1}{|\mathcal{V}||\mathcal{A}|}\sum_{(i,a)\in\mathcal{V}\times\mathcal{A}} D_{\mathrm{KL}}\left(p(j\mid i)\,||\,q^{a,b}_\phi(j\mid i)\right)\right] = \underbrace{\mathbb{E}_{a,b\sim\mathrm{U}_\mathcal{A}}\left[\mathrm{Loss}^{a,b}_{\mathrm{Node}}\right]}_{\text{Node-level loss under } \mathrm{U}_\mathcal{A}}$$

Augmentation-level similarity

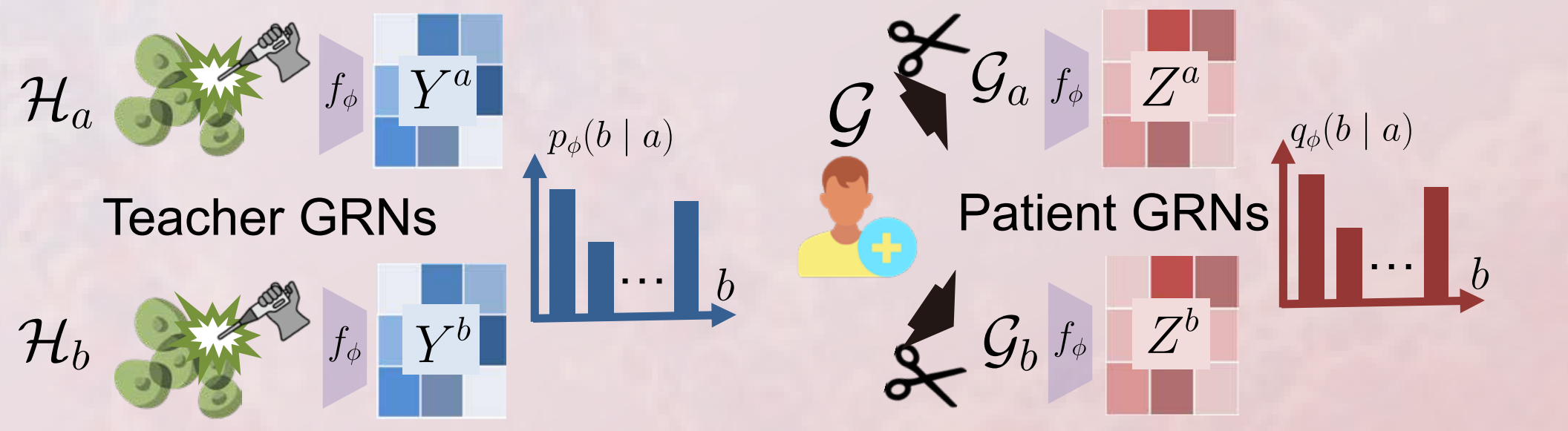
$Z^a \triangleq f_\phi(\mathcal{G}_a) \in \mathbb{R}^{|\mathcal{V}|\times d}$
$Y^a \triangleq f_\phi(\mathcal{H}_a) \in \mathbb{R}^{|\mathcal{V}|\times d}$

p_ϕ : Biological teacher distribution

$$p_\phi(b\mid a) \triangleq \frac{\exp\left(\mathrm{sim}_F(Y^a, Y^b)/\tau_{\mathrm{a}}\right)}{\sum_{c\in\mathcal{A}}\exp\left(\mathrm{sim}_F(Y^a, Y^c)/\tau_{\mathrm{a}}\right)}$$

q_ϕ : Learned distribution

$$q_\phi(b\mid a) \triangleq \frac{\exp\left(\mathrm{sim}_F(Z^a, Z^b)/\tau_{\mathrm{a}}\right)}{\sum_{c\in\mathcal{A}}\exp\left(\mathrm{sim}_F(Z^a, Z^c)/\tau_{\mathrm{a}}\right)}$$



Proposed method

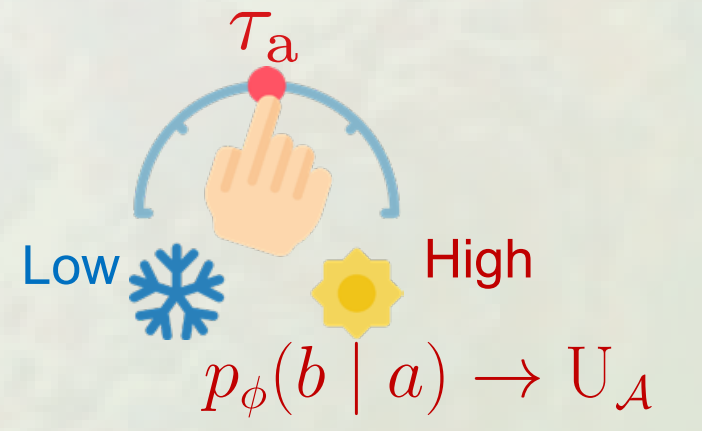
Theorem 4.1

Under the assumption: $p_\phi(i,j,a,b) = p(i,j)p_\phi(a,b)$

$$\mathrm{Loss}_{\mathrm{SupGCL}} = \underbrace{\mathbb{E}_{a,b\sim p_\phi(b|a)\mathrm{U}_\mathcal{A}(a)}\left[\mathrm{Loss}^{a,b}_{\mathrm{Node}}\right]}_{} + \underbrace{\frac{1}{|\mathcal{A}|}\sum_{a\in\mathcal{A}} D_{\mathrm{KL}}\left(p_\phi(b\mid a)\,||\,q_\phi(b\mid a)\right)}_{}$$

1. Node-level loss under $p_\phi(b\mid a)$
   - Local node-level learning of gene-specific consistency
2. Augmentation-level loss
   - Global augmentation-level learning of graph-wide perturbations

Corollary 4.2: Relation to existing GCL



$p_\phi(b\mid a) \to \mathrm{U}_\mathcal{A}$

$$\lim_{\tau_{\mathrm{a}}\to\infty} \mathrm{Loss}_{\mathrm{SupGCL}} = \mathrm{Loss}_{\mathrm{Node}}$$

τ_{a} continuously controls the degree to which teacher information is reflected

★ Node-level GCL naturally emerges as a special case $(\tau_{\mathrm{a}}\to\infty)$

Experiments

Data

TCGA: Construct patient-specific GRNs from TCGA cancer patient expression profiles

Shared genes: 975

LINCS: Construct teacher GRNs from LINCS L1000 knockdown cancer cell line expression profiles

Node feature = gene expression
Edge feature = edge contribution value (weighted directed graph)

3 cancer types:

Breast
Lung
Colorectal

Model

★ 5-layer Graph Transformer
★ Pretrain, then fine-tune on downstream tasks

Pretraining Baselines:

- w/o pretrain
- GAE[2]
- GraphCL[3]
- GRACE[4]
- SGRL[5] (SOTA)

Ours :

- SupGCL

Downstream tasks

Node-level tasks

* BP : Biological Process (based on Gene Ontology)

- BP. classification*
  - Metabolic process
  - Signal transduction
  - Cellular organization

Graph-level tasks

High risk

- Hazard prediction
- Subtype classification (Breast - only)

Results

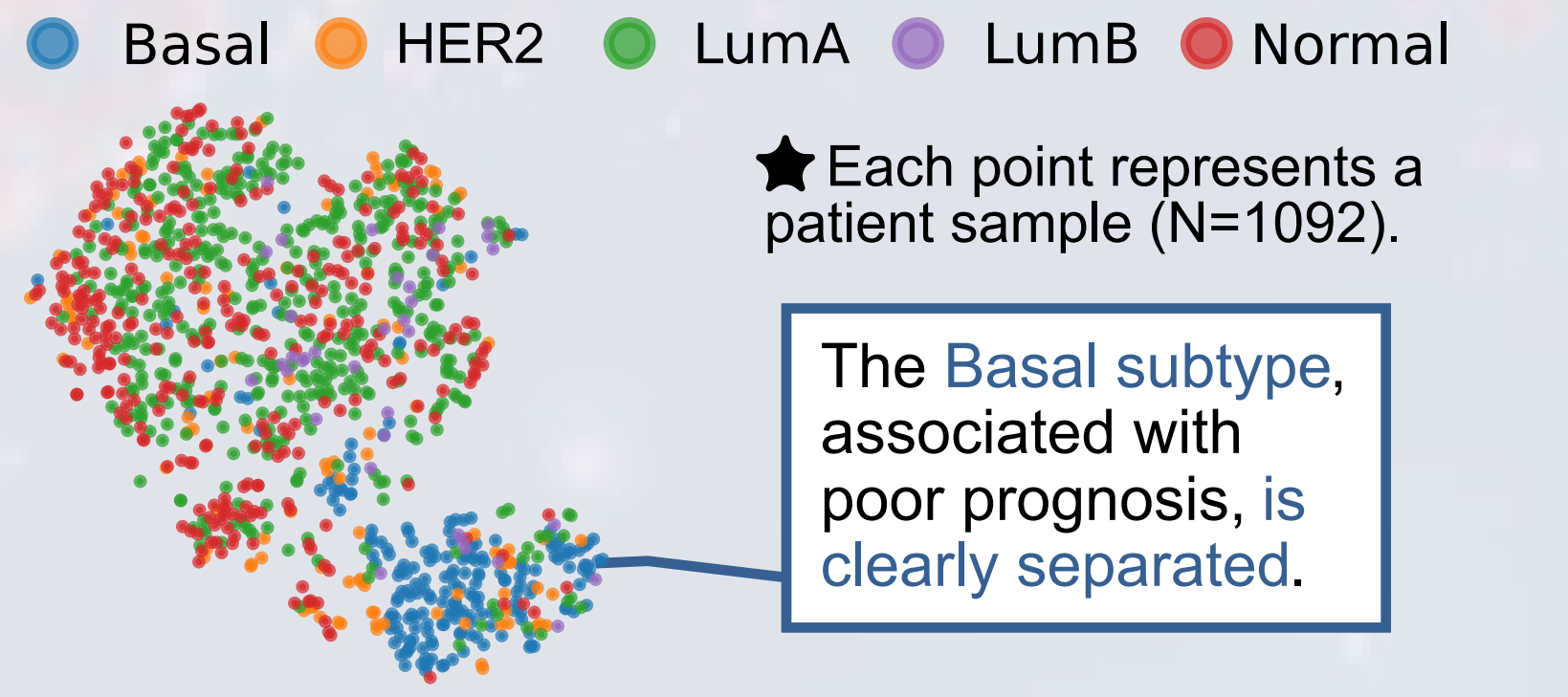
Zero-shot evaluation

Breast cancer subtype clustering performance of pretrained embeddings

Zero-shot clustering	GAE	GraphCL	GRACE	SGRL	SupGCL
Subtype					
Breast (ARI)	0.020 ± 0.000	0.050 ± 0.020	0.107 ± 0.001	0.071 ± 0.025	**0.121 ± 0.007**
Breast (NMI)	0.029 ± 0.001	0.081 ± 0.027	0.132 ± 0.017	0.092 ± 0.033	**0.155 ± 0.013**

✓ SupGCL achieved the clearest subtype separation among all methods.

This suggests that SupGCL learns subtype-specific network representations.



★ Each point represents a patient sample (N=1092).

The Basal subtype, associated with poor prognosis, is clearly separated.

Downstream task evaluation

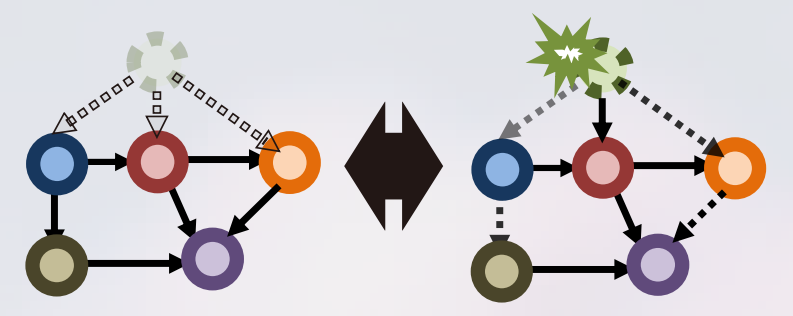
Fine-tuning performance for each task across all cancer types

Evaluation metrics BP: Subset Accuracy, Hazard: C-index, Subtype: Accuracy

	Task	w/o-pretrain	GAE	GraphCL	GRACE	SGRL	SupGCL
	BP.						
Node	Breast	0.232 ± 0.031	0.230 ± 0.029	0.167 ± 0.042	0.230 ± 0.051	0.220 ± 0.052	**0.243 ± 0.052**
	Lung	0.259 ± 0.056	0.247 ± 0.038	0.115 ± 0.024	0.259 ± 0.063	0.233 ± 0.027	**0.282 ± 0.037**
	Colorectal	0.231 ± 0.062	0.245 ± 0.023	0.207 ± 0.058	0.249 ± 0.050	0.146 ± 0.029	**0.262 ± 0.030**
	Hazard						
Graph	Breast	0.601 ± 0.035	0.625 ± 0.035	0.638 ± 0.049	0.642 ± 0.064	0.640 ± 0.077	**0.650 ± 0.059**
	Lung	0.611 ± 0.052	0.619 ± 0.062	0.616 ± 0.049	0.609 ± 0.055	0.611 ± 0.060	**0.627 ± 0.051**
	Colorectal	0.621 ± 0.070	0.631 ± 0.091	0.657 ± 0.071	0.647 ± 0.059	0.616 ± 0.123	**0.698 ± 0.085**
	Subtype						
	Breast	0.804 ± 0.031	0.834 ± 0.028	0.719 ± 0.077	0.841 ± 0.026	0.829 ± 0.030	**0.847 ± 0.036**

✓ SupGCL consistently outperformed baseline methods on both node-level and graph-level tasks.

Conclusion



- **SupGCL** — a supervised, probabilistic extension of conventional GCL.
- Real knockdown experiments as biologically grounded augmentation supervision.
- Improved representations and downstream performance across all tasks on clinical data.
- Future: scalable pan-cancer pretraining and cross-domain generalization.

Reference

Code

Paper

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